

GENIE

Google Gemini Powered Learning Assistant

Requirement Analysis Phase

Identifying functional, technical, and user requirements

Project Phase Documentation

This document covers the activities, outputs, and considerations for this phase of the Genie project.

1. Functional Requirements

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- The application should accept questions or learning-related prompts from the user.
- It should send the prompt to the configured AI service and receive a generated response.
- It should display the response in a readable format.
- The assistant should support explanations, summaries, examples, and general study-related questions.
- The system should provide clear feedback when an input or service request cannot be processed.

Aspect	Details
Purpose	Identifying functional, technical, and user requirements
Project	Genie: Google Gemini Powered Learning Assistant
Technology	Python + Google Gemini integration
Target Users	Students / learners

2. Technical & Software Requirements

Genie – Requirement Analysis Phase

- Programming language: Python.
- AI service/model: Google Gemini API or an approved Gemini model configuration.
- Development environment: VS Code or another Python-compatible IDE.
- Required Python packages depend on the selected implementation and Gemini SDK/API method.
- Internet connectivity is required when the application communicates with the cloud-based AI service.

Key Note: The exact implementation can be adjusted based on the final project code and available Gemini API configuration.

3. User & Quality Requirements

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- The interface should be simple and easy for students to use.
- Responses should be understandable and relevant to the submitted question.
- The application should handle common input errors gracefully.
- API credentials should be kept secure and should not be hard-coded into publicly shared source code.
- The project should be maintainable, testable, and suitable for demonstration.

Phase Output: A clear, reviewable result that can be carried forward to the next project phase.